Technical Note — EV + Developer Engine (5-Year Default)

1) Background & Purpose

This engine distributes project probability across candidate FID years, computes expected values by year, and evaluates developer economics (DevCost, MOIC, k*, IRR). It is a post-scoring tool, not a re-fit of Bayes or Cox models.

2) Inputs

- Required: blended_prob (or p_bayes/p_cox), project_cost, years_remaining, urgency_scale_(0-1)
- Optional: imputed_cost, start_year, end_year

3) Methodology

EV Allocation:

Candidate years = BASE_YEAR+1 .. BASE_YEAR+horizon. Annual probabilities allocated with softmax weighting (alpha0, alpha1) using urgency and years_remaining. EV per year = P(year) × project_cost.

Developer Engine:

Cost basis selected from project_cost → imputed_cost → value_proxy. DevCost = dev_cost_pct × cost_basis. Expected FID year = weighted average over candidate years. PV sums computed using S-curve spend paths (S, even, front, back). Metrics include k_star, PV_MOIC_k, IRR_k, k_star_adj.

4) Outputs

- Annual_p_YYYY, EV_YYYY columns
- EV_cum, EV_disc
- Developer metrics: DevCost, E_FID_year, PV_MOIC_k, k_star, IRR_k, k_star_adj

5) Replication

- Generate input with MPI Risk Engine
- Run EV + Developer Engine with chosen knobs
- Archive inputs/outputs
- Record command/config for audit

6) QA Checks

- Σ annual_p $\approx$ total probability
- EV_disc $\leq$ EV_cum when discount > 0
- S-curve weights sum to 1

- IRR_k finite and reasonable

7) Limitations

Stylized, counterfactual parameters; not calibrated to real projects. Results show marginal and distributional effects, not forecasts.